Exercise 3.6 (*)

Suppose $f : (a, b) \to \mathbb{R}$ is uniformly continuous, and $(x_j)_{j=0}^{\infty}, (y_j)_{j=0}^{\infty} \subset (a, b)$ satisfy $x_j \to a$ and $y_j \to b$.

(a) Show that the sequences $(f(x_j))_{j=0}^{\infty}$ and $(f(y_j))_{j=0}^{\infty}$ are Cauchy.

Solution. Given $\varepsilon > 0$ choose $\delta > 0$ such that $|u - v| < \delta \Rightarrow |f(u) - f(v)| < \varepsilon$ (uniform continuity). Since $x_j \to a$ and $y_j \to b$, both (x_j) and (y_j) are Cauchy. Hence there is N with $|x_j - x_k| < \delta$ and $|y_j - y_k| < \delta$ for all $j, k \geq N$, which implies $|f(x_j) - f(x_k)| < \varepsilon$ and $|f(y_j) - f(y_k)| < \varepsilon$ for $j, k \geq N$. Thus both image sequences are Cauchy.

(b) Setting $f_L := \lim_{j\to\infty} f(x_j)$ and $f_R := \lim_{j\to\infty} f(y_j)$, show that

$$\tilde{f} : [a, b] \to \mathbb{R}, \qquad \tilde{f}(x) = \begin{cases} f_L, & x = a, \\ f(x), & a < x < b, \\ f_R, & x = b, \end{cases}$$

is continuous on $[a, b]$.

Solution. By (a) the limits f_L, f_R exist (Cauchy in $\mathbb{R}$ hence convergent). On (a, b), $\tilde{f} = f$, so continuity holds. We check continuity at a; the case b is analogous.

Let $\varepsilon > 0$. Pick $\delta > 0$ for uniform continuity of f and choose J large so that $|x_J - a| < \delta/2$ and $|f(x_J) - f_L| < \varepsilon$. If $0 < |t - a| < \delta/2$, then $|t - x_J| \leq |t - a| + |x_J - a| < \delta$, hence

$$|\tilde{f}(t) - \tilde{f}(a)| = |f(t) - f_L| \leq |f(t) - f(x_J)| + |f(x_J) - f_L| < \varepsilon + \varepsilon = 2\varepsilon.$$

Therefore $\tilde{f}$ is continuous at a. A symmetric argument using (y_j) gives continuity at b. Thus $\tilde{f}$ is continuous on $[a, b]$.

(c) Deduce that f is uniformly continuous on (a, b) if and only if there exists a continuous $\tilde{f} : [a, b] \to \mathbb{R}$ with $\tilde{f}(x) = f(x)$ for all $x \in (a, b)$ (i.e. $\tilde{f}$ is an extension of f).

Solution. ($\Rightarrow$) If f is uniformly continuous, parts (a)–(b) construct a continuous extension $\tilde{f}$ on $[a, b]$.

($\Leftarrow$) If such $\tilde{f}$ exists, then $\tilde{f}$ is continuous on the compact set $[a, b]$, hence uniformly continuous. Its restriction to (a, b) equals f and is therefore uniformly continuous.